Time-Series Data Encodes Dynamics, Not Independent Observations

Non-stationarity, dependence, dimensionality, and non-ergodicity in real-world data

- Different time-series structures (univariate, multivariate, panel, mixed-frequency) require different tools
- Time-series are powerful because they capture evolution, but fragile because real data rarely behaves ideally

Beyond i.i.d.: The Reality of Time-Series Data

TS captures real-world dynamics, Natural ordering, Feature engineering, Supports advanced sequential model

Some Challenges

- Non-stationarity, Levels/trends change; Autocorrelation violates **i.i.d. assumptions** - i.i.d. assumes no memory and no change in distribution
- Irregular or mixed-frequency observations
- Noise, jumps, fat tails
- Curse of dimensionality (multivariate cases)
- Large cross-sections → unstable covariance matrices → need shrinkage, PCA/DFM, embedding

Impacts

- Time-series requires specialized methods……

Structural and Statistical Challenges in Time-Series Data

1. Non-Stationarity & Structural Breaks

- Volatility clusters; regimes shift, structural breaks & changing distributions
- Models trained in one regime often fail in another

2. Autocorrelation & Dependence Violates ML Assumptions

- Many ML models assume i.i.d. unless preprocessed properly - i.i.d. assumes no memory and no change in distribution
  - Almost never true.
    - ✓ Volatility changes (GARCH effect)
    - ✓ Means drift (trend)
    - ✓ Regimes change (bull/bear)

3. Irregular, Mixed-Frequency, or Missing Observations

- Daily vs monthly vs quarterly alignment
- Non-trading days, revisions, ragged edges
- Temporal aggregation: attenuate correlations, create moving-average behaviour artificially, smooth away turning points

4. Noise, Outliers, and "Market Shocks"

- Fat tails and jumps distort estimators
- Sudden discontinuities break smooth assumptions

5. High-Dimensional When Multivariate

- Covariance instability
- Curse of dimensionality
- Requires shrinkage, factor models, or embedding methods

Fundamental Limits of Time-Series Modeling

6. Ergodicity vs Non-Ergodicity

- **Ergodic assumption:** *The future resembles the past; long-run averages converge to expectations, m*ost forecasting assumes this implicitly
- **Most economic and financial series are non-ergodic**

7. The "Shortest Series Problem"

- Multivariate models often require **aligned series of equal length**

8. Curse of Dimensionality

- With **N variables**, covariance matrix requires **N(N−1)/2** unique covariances
- Example: 10 variables → 45 covariances, 50 variables → 1,225 covariances

9. Spurious Correlations

- With many variables but limited data : random noise can look like meaningful relationships

10. Seasonality (Deterministic vs Stochastic)

11. Nonlinear Dynamics & Threshold Behaviour

- Regime-dependent autocorrelation **,** Volatility clustering (ARCH/GARCH), Markov switching

12. Ex-post Data Revisions (Macroeconomics)

- Impact backtesting

13. Scaling & Normalization Issues

- Standardization changes factor loadings, Some transformations preserve trends, others destroy them, Log-levels vs percent-change decisions

14. Multicollinearity

- Variable predictors correlations **through time,** rolling VIF shows **unstable coefficients**

Why Stationarity Is Assumed — and What It Costs

Why Stationarity Is Assumed

- Non-stationary series have time-varying means, variances, or dynamics – most statistical tests assume stable distributions of data

- Many classical (some modern) models assume stationarity; else results can be misleading (spurious regressions):
  - **Classical**
    - AR, MA, ARMA
    - ARIMA (after differencing)
    - VAR, VECM
    - State-space models with stationary innovations
  - **Modern / ML**
    - Regime-switching, HMM (PS)
    - Kalman filters (NS)
    - LSTMs and transformers (NA)
    - Etc.

S – ASSUNES STATIONATY
PS – PIECEWISE STATIONARY
NS – HANDLES NON-STATIONARY
NA – NO EXPLIXIT ASSUMPTION BUT SENSITIVE TO DRIFT

- Segmentation, motifs, and clustering require homogeneous regimes

However, there are a family of models that can handle non-stationary data e.g.

- Dynamic Linear Models (DLM) (NS)
- Vector Error Correction Models(VECM) (NS)
- Bayesian structural models (NS)
- Kalman-smoother trend decomposition (NS, **smoothing is two-sided / ex-post**)
- Other…

Stationarity comes at a cost: you lose trend, structure, and long-run information

Stationarity as an Engineering Constraint, Not a Truth – Model Ready

LOG PRICES

Input	Formalism	Use Case	Not Suitable
Log(price) (Close, Open, High, Low)	$P_t, \ln(P_t)$	Trend extraction (HP, SSA), long-term segmentation	Econometrics (ARIMA/VAR), regressions, ML forecasting

SIMPLE RETURNS

Input	Formalism	Use Case	Not Suitable
Percent returns	$r_t = \frac{P_t}{P_{t-1}} - 1$	Portfolio math, correlations, risk metrics, ML features	Time-additive models, volatility modelling

PERCENT CHANGE RETURNS

Input	Formalism	Use Case	Not Suitable
pct_change()	$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$	Reporting, intuitive %-moves	GARCH/ARMA/VAR, additive models

LOG RETURNS

Input	Formalism	Use Case	Not Suitable
Log-diff	$r_t = \ln(P_t) - \ln(P_{t-1})$	ARMA/VAR, GARCH, ML forecasting, segmentation, regime classification	%-based interpretation

RANKING

Input	Formalism	Use Case	Not Suitable
Any stationary feature	$u_t = \frac{R_t}{N}$	Robust ML features, **regime classification**	ARIMA/VAR, GARCH, cointegration

How Macroeconomic Time Series Are Made Econometrically Tractable

Transform / Input	Formalism	Typical econometric use	Not suitable
Log-diff (1-period growth)	$\Delta \ln X_t = \ln X_t - \ln X_{t-1}$	Standard for **TS econometrics**: VAR, ARIMA, Granger, local projections when modeling **short-run dynamics** of trending macro series (GDP, INDPRO, CPI, money, credit).	Series already stationary (rates, spreads, indices) or where trend is already removed.
Percent change	$\%\Delta X_t = \frac{X_t - X_{t-1}}{X_{t-1}}$	Very common in **applied / micro econometrics**: growth regressions, elasticities, panel models, DiD, cross-country growth; also fine in time-series regressions as long as stationarity is ok.	When you need clean time-aggregation & variance stabilization (log-diff is better for strict TS dynamics).
YoY growth	$\Delta_{12} \ln X_t = \ln X_t - \ln X_{t-12}$	Used a lot in **policy / applied macro** (YoY GDP growth, YoY CPI), long-horizon growth / inflation, smoother regressors in LP/DSGE, low-freq VARs.	High-frequency VAR / impulse response work where you care about month-to-month dynamics (YoY overlaps and induces MA structure).
Log-level	$\ln X_t$	Input for **trend/cycle decomposition** (HP, SSA, Baxter-King), sometimes in cointegration setups.	Direct VAR/ARIMA on raw logs for trending macro aggregates without cointegration / trend handling.
Raw levels	X_t	Only for variables that are **plausibly stationary in level**: many spreads, some indices, some rates after allowing for breaks.	Trending aggregates like GDP, CPI, INDPRO, money, credit.

Linear Structure Enables Econometric Inference, Not Necessarily Prediction

Linear data

- Residual diagnostics apply only to classic econometric models, while performance metrics like RMSE, MAE, MAPE, and even R^2 apply universally to both econometric and modern ML/DL models — regardless of whether the data is stationary or not.

- Linear models depend on linear, stationary structure; ML/DL models learn nonlinear patterns and handle nonstationary directly.

Nonlinearity: Essential for Residuals, Irrelevant for Prediction Metrics

Non-linear data

- Non-linear data—univariate or multivariate—invalidates econometric residual assumptions but has zero impact on prediction metrics like RMSE/MAE/MAPE, etc.
- Non-linear models often bypass the need for stationarity entirely e.g. Random Forest, Gradient Boosting (XGBoost, LightGBM), LSTM / GRU, Transformers etc.
- **Key1 :** diagnose the data
- **Key 2:** Set objective trade-off between **structural interpretation or predictive accuracy**

Non-linear relationships

- Detection of non-linear relationships is achieved with plots & tests
- Without addressing this, residuals will not be random e.g. patterns, curvature etc
- Approach using transformations, splines, or ML/ DL models
- Key 1 : Trade-off between intention : interpretability or prediction

So what?

- **Markets are nonlinear**: linear models can be misspecified even if RMSE/MAE/MAPE look OK
- **Econometrics prioritizes identification/interpretation** (linearity is a convenient assumption)
- **The choice is strategic**: interpretation vs prediction
- **Nonlinearity mainly breaks interpretation**, not point-forecast accuracy

Diagnosing and Modelling Nonlinearity

Non-linear data - detect it

- Use **plots**: scatterplots, partial residuals, smoothers (LOESS), GAM plots
- Use **tests**: Ramsey RESET (misspecification), BDS (serial nonlinearity), ACF diagnostics
- Key: **Don't force stationarity unless required by the model**

Non-linear relationships

- between **factors -> target**
  - ✓ **Plots**: partial residuals, smoothers (LOESS), GAM curves
  - ✓ **Tests**: Ramsey RESET, component-plus-residual plots
  - ✓ **Spline models** (GAM): interpretable smooth nonlinearity

- between the **factors**
  - ✓ **Scatterplots & pairwise smoothers**
  - ✓ **Correlation instability across regimes**
  - ✓ **Non-linear dependence tests** (BDS, mutual information, HSIC)
  - ✓ **Model interactions explicitly**: ($X_1 \times X_2$), tensor GAMs

Feature engineering can create new nonlinear relationships e.g. windows/smoothing, rolling vol/corr, moving averages, z-scores, regime indicators etc. – many of these types can be non-linear transformations